

The Secretary

Approved/Not Approved

 13/11/25

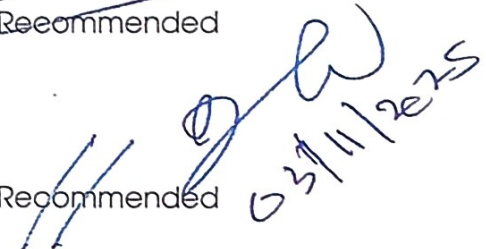
Chief Director, Water Resources
& Irrigation Development

Recommended/Not Recommended

 01/11/2025

Director ICT

Recommended/Not Recommended

 03/11/2025

Director Irrigation Development

Recommended/Not Recommended

 03/11/2025

**REF: REQUEST FOR APPROVAL OF THE NATIONAL IRRIGATION GEOSPATIAL DATABASE
CONCEPT**

The National Irrigation Database Concept Document

The National Irrigation Database is a strategic initiative aimed at improving agricultural productivity and resilience through the establishment of a centralized, secure, and data-driven platform for the collection, management, and visualization of irrigation-related information across Zimbabwe. The database will integrate data from both public and private irrigation initiatives, providing a comprehensive national picture of irrigation activities and infrastructure.

Leveraging advanced geospatial and satellite technologies, the platform will facilitate real-time monitoring of irrigation infrastructure, track operational status and maintenance needs, and support agricultural output analysis including yield forecasting. Additionally, it will generate early warning alerts for irrigation-related disasters such as flooding in low-lying areas or frost occurrences, thereby enhancing preparedness and mitigation efforts. These technological improvements will enable continuous assessment of irrigation performance, better resource allocation, and more responsive planning across the agricultural sector.

Attached is the concept note for your approval

Return file: Director Irrigation Development

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The system is designed to promote evidence-based decision-making by providing accurate, up-to-date, and location-specific data to policymakers, researchers, and other stakeholders. This will contribute significantly to the achievement of national goals on food security, sustainable agriculture, and rural development, in alignment with Zimbabwe's Vision 2030.

Objectives

- 1. Develop a Centralized Data Repository:** Create a unified platform for collecting, storing, and updating irrigation data from all provinces, capturing information on infrastructure, water sources, land use, crop types, and productivity levels.
- 2. Enhance Monitoring and Evaluation Capabilities:** Utilize geospatial, remote sensing, and satellite technologies to provide real-time tracking of irrigation infrastructure status, operational performance, and maintenance requirements, enabling timely interventions and efficient resource management.
- 3. Facilitate Early Warning and Disaster Preparedness:** Integrate predictive tools and alert systems to detect irrigation-related hazards such as floods, droughts, or frost, supporting proactive response and minimizing production losses.

Quantity	Item	Cost
1	GIS Mobile Application Development	\$3,000
1	Training of Trainers and users	\$5,000
1	Server Configuration	\$1,200
1	Contingency	\$7,000
1	Hosting per annum	\$500
Total		50,700

Conclusion

The National Irrigation Database will significantly contribute to improving agricultural practices in Zimbabwe by providing accurate and timely data. This initiative will foster better decision-making and enhance food security for rural communities.

4. Strengthen Policy Formulation and Planning: Provide accurate, comprehensive, and timely data to inform government policies, investment decisions, and development strategies in the agricultural and water management sectors.

5. Promote Collaboration and Data Sharing: Foster partnerships among key stakeholders—including government ministries, research institutions, development partners, and private sector actors—through a shared data platform that enhances coordination and accountability.

6. Support Zimbabwe's Vision 2030: Contribute to the national agenda of achieving an upper-middle-income economy by 2030 through improved agricultural productivity, sustainable water use, and enhanced rural livelihoods driven by technology and innovation.

Key Components of proposed database

4. Mobile Application

- **Data Collection:** Users can input data regarding irrigation schemes (e.g., plot size, crop types, water usage) through the app.
- **User Interface:** An intuitive interface will facilitate easy data entry related to irrigation status and infrastructure.
- **Offline Capability:** The mobile app will function without internet connectivity, syncing data once online.

2. Server/Database

- **Backend Framework:** Developed using Django, ensuring robust management of business logic and data processing.
- **Database:** PostgreSQL with PostGIS will store spatial data and manage attributes related to irrigation schemes.
- **API Development:** RESTful APIs will enable data exchange between the mobile app and the server.

3. Geospatial Components

- **GeoDjango:** To manage geographic data, enabling spatial queries and analytics.
- **GeoServer:** For serving geospatial data and generating visual map views of irrigation schemes.

4. Frontend Interface (WebGIS Application)

- **Map View:** Utilizing Leaflet.js, the map will display irrigation scheme locations, with popups showing attributes such as: